Start Here

SUBStart Here

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Meeting folder

Everything needed to discuss this work, in one place.

Read in this order

Everything is a PDF. The `.md` files beside them are the editable sources; run `python make_pdfs.py` to rebuild.

#	file	what it is	time
0	`00-plain-english.pdf`	The whole thing with no jargon, plus a glossary	6 min
1	`02-findings.pdf`	The findings, with the numbers	12 min
2	`03-the-ask.pdf`	The one decision I need from you	2 min
3	`papers/01-mittelstadt-2023-*.pdf`, section 6 only	The closest competing work	15 min
4	`04-paper-draft.pdf`	The full paper draft, in academic style	40 min
5	`05-reading-notes.pdf`	What I checked in each cited paper, and what changed	10 min

`01-start-here.pdf` is this document. **If any wording is unfamiliar, start at 00** — it defines every term the others use.

In one sentence

> These fairness fixes are supposed to make two groups equal. They can do that either by approving more people from the group being turned down, or by rejecting more from the group getting through — and the score that certifies the fix cannot tell which happened. I found that which one you get depends on how generous the system already is, and almost every real system is the kind where you get the bad one.

The papers, and why each is here

file	why it matters
01 Mittelstadt et al. (2023)	**The closest competing work.** Showed these fixes mostly harm rather than help, *and* proposed the remedy I thought was mine. Read section 6.
02 Corbett-Davies et al. (2017)	Proves what the mathematically best possible fair program looks like — which makes one of my comparisons a ceiling rather than a rival.
03 Diana et al. (2021)	The alternative method the field recommends for this problem. I test it and it performs badly.
04 Kearns et al. (2018)	Showed fixes can look fine per group while failing for combinations of groups. I reproduce this.
05 Goethals et al. (2024)	**The near miss.** Studies the same dimension I do, calls it "overlooked" — and two of its four authors also wrote paper 01. Their setup fixes the number of approvals in advance, so my effect cannot appear in it.
06 Menon & Williamson (2018)	Supporting theory for paper 02.
07 Ustun et al. (2019)	Another "avoid harm" method from the literature.
08 Black et al. (2022)	On models that score equally well but disagree about individuals. Supports one of my cautions.
09 Agarwal et al. (2018)	The method everything here is built on.
10 Ding et al. (2021)	Argues the field over-relies on one old dataset, and supplies replacements. I use them.

Where the underlying work is

- `../docs/11–26` — 16 documents, one per finding, each stating its own limits
- `../../docs/01–10` — the coursework this grew out of
- `../results/` — every number as a spreadsheet
- `../../tests/test_documented_claims.py` — regenerates every headline figure and fails if a document disagrees with its data
- `git log` — the full record, including predictions committed before the experiments that tested them

Two things I would rather say than be asked

1. My remedy is not new. I believed the fix I proposed was original. It is a variation on one already published (paper 01, section 6). I found this myself by reading their paper, corrected my write-up, and demoted it to a supporting result. What survives: mine works without the program ever seeing which group someone belongs to, and it is tested far more widely.

2. My explanation failed. I tried to derive *why* the flip happens, wrote the prediction down in advance, and tested it on new data. It passed my own test and was then beaten by a much cruder rule, so it explains nothing. It is written up as a failure. I can say when the flip happens, not why.

Scale

26 datasets · 61 experimental runs · 3 data sources · 2 kinds of decision · 5 repeats of everything